

Lab 2: [replace with the lab title]

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Setup

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

Part A: [rename to match the lab]

Write one sentence describing what this part is doing.

```
def decay_numerical(C0, k, dt, t_end):
    """Return arrays of time and concentration for forward-Euler tracer decay."""
    n = int(t_end / dt)
    C = C0
    ts = [0.0]
    Cs = [C0]
    for i in range(n):
        C = C + (-k * C) * dt          # forward difference
        ts.append((i + 1) * dt)
        Cs.append(C)
    return np.array(ts), np.array(Cs)

# The exact answer, for comparison
C0, k, t_end = 100.0, 1.0, 24.0
t_fine = np.linspace(0, t_end, 200)
exact = C0 * np.exp(-k * t_fine)

fig, ax = plt.subplots(figsize=(7, 4))
```

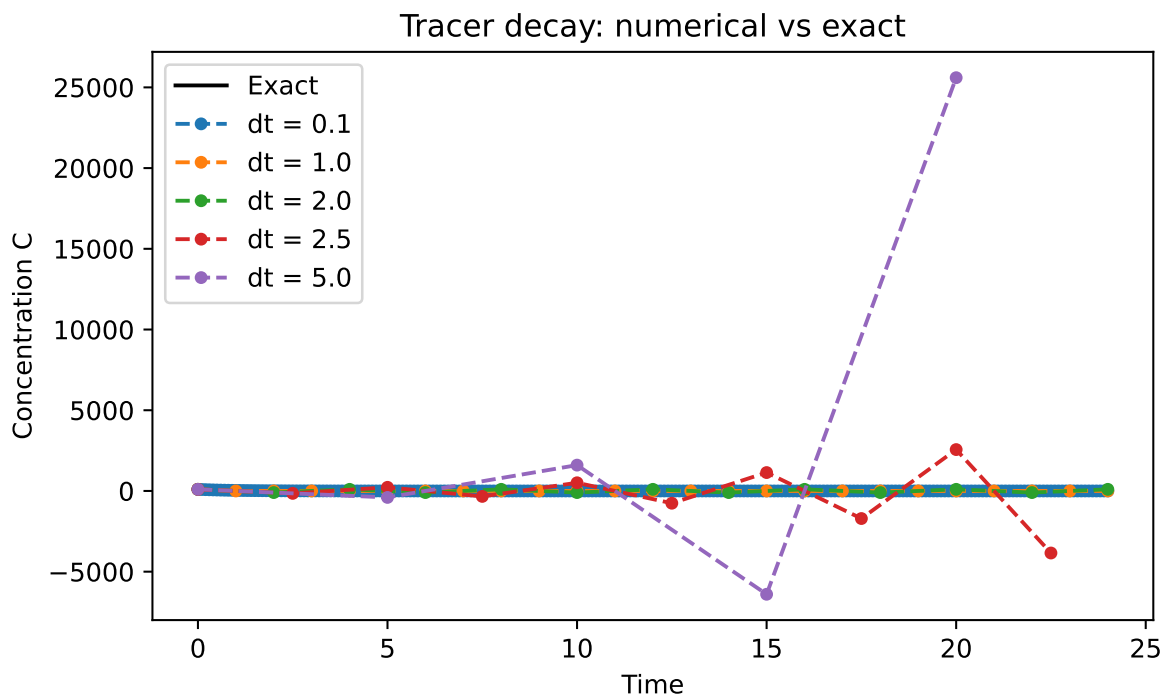
```

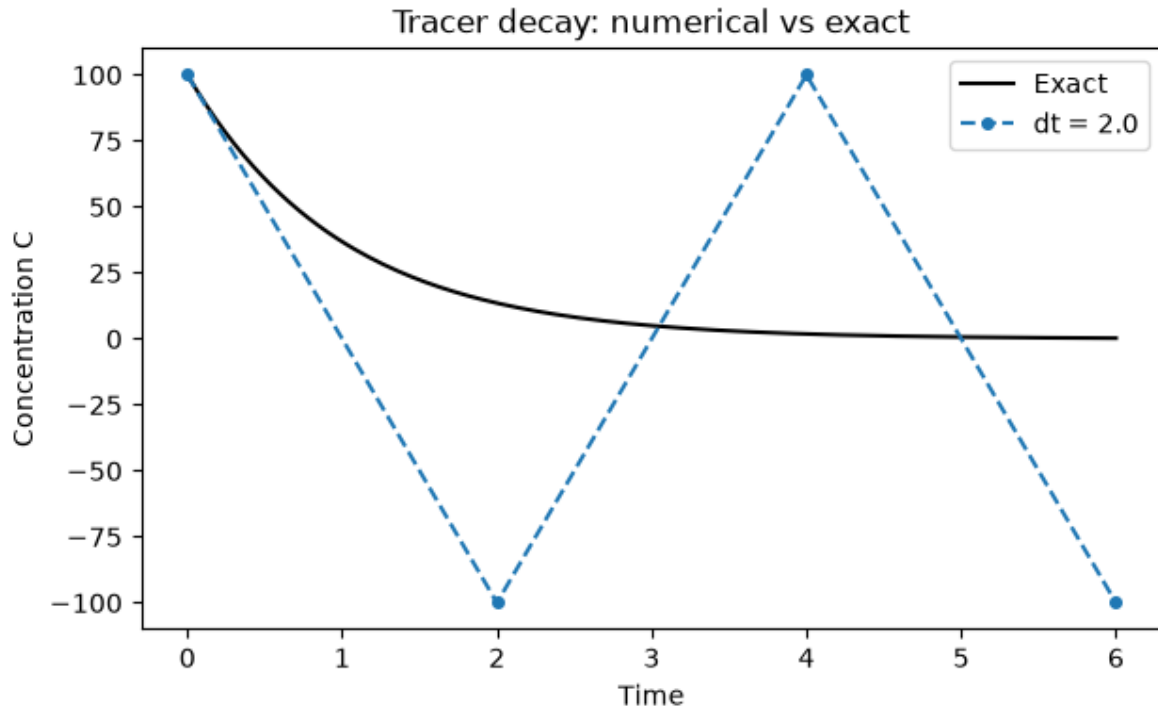
ax.plot(t_fine, exact, "k-", label="Exact")

for dt in [0.1, 1.0, 2.0, 2.5, 5.0]:
    ts, Cs = decay_numerical(C0, k, dt, t_end)
    ax.plot(ts, Cs, "o--", markersize=4, label=f"dt = {dt}")

ax.set_xlabel("Time")
ax.set_ylabel("Concentration C")
ax.set_title("Tracer decay: numerical vs exact")
ax.legend()
plt.show()
# The e

```





Answer to task A1: (one to three sentences) k rate controls the gradient of the curve.

Answer to task A2: (one to three sentences) 2

Answer to task A3: (one to three sentences) 1

Answer to task A4: (one to three sentences) When dt is large, the numerical solution becomes a negative due to the equation multiplying the value with the C old value, the larger the dt number the larger the negative. This is then over compensated and shoots positive. the larger the dt value, the larger the oscillation. ## Part B: [rename to match the lab]

```
import numpy as np
import matplotlib.pyplot as plt

# ---- Parameters ----
V = 1.0e6          # lake volume (m3)
Q = 5.0e3          # river inflow (m3 per day)
C_in = 10.0        # incoming concentration (mg/L)
k = 0.05           # in-lake decay rate (per day)

# ---- Time settings ----
dt = 1.0           # time step (days)
days = 200

# ---- Initial condition ----
C = 0.0            # clean lake
```

```

# ---- Storage ----
history = [C]

# ---- Time loop ----
for _ in range(days):
    dCdt = (Q * C_in - Q * C - k * V * C) / V # the rate of change
    C = C + dCdt * dt                         # forward difference
    history.append(C)

# ---- A quick look at the numbers ----
print(f"Started at {history[0]:.2f} mg/L")
print(f"After 10 days: {history[10]:.2f} mg/L")
print(f"After 50 days: {history[50]:.2f} mg/L")
print(f"After {days} days: {history[-1]:.2f} mg/L")

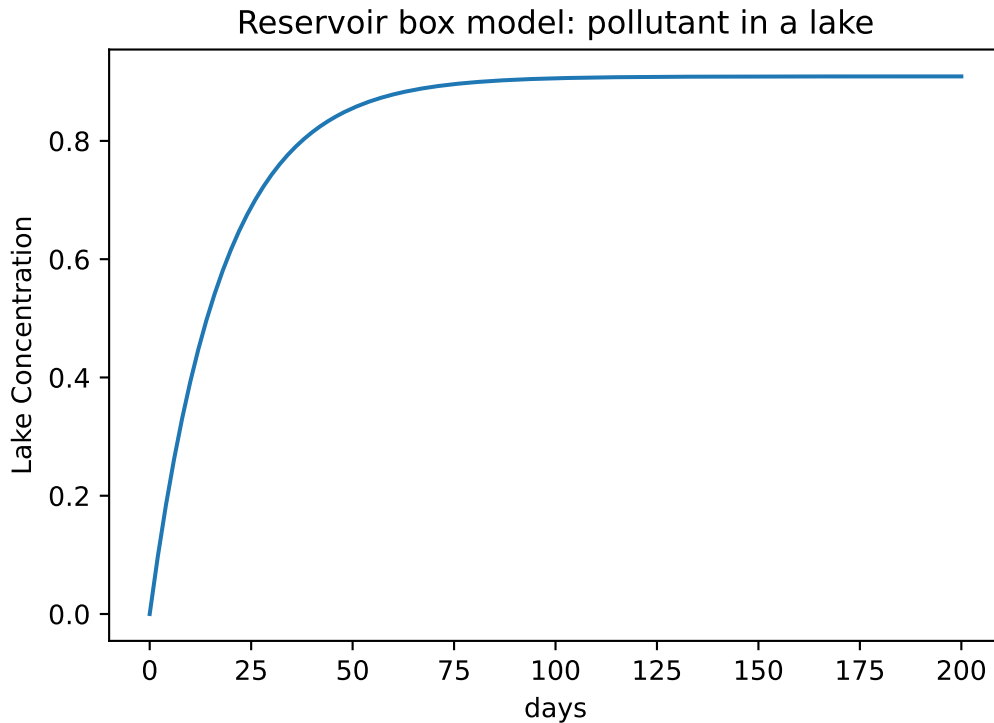
fig, ax = plt.subplots(figsize=(6, 4))
ax.plot(range(days + 1), history)
ax.set_xlabel("days")
ax.set_ylabel("Lake Concentration")
ax.set_title("Reservoir box model: pollutant in a lake")
plt.show()

```

```

Started at 0.00 mg/L
After 10 days: 0.39 mg/L
After 50 days: 0.86 mg/L
After 200 days: 0.91 mg/L

```



Answer to task B1: *(short answer)*

Reflection

Write two to four sentences on what surprised you, what you got stuck on, and what you would do differently next time. This section is worth marks — do not leave it blank.

Checklist before you push

- ☐ Every task has a code cell **and** a short prose answer
- ☐ Every figure has axis labels and a title
- ☐ Your name and student ID are in the YAML header
- ☐ The file renders cleanly (no red error boxes in the HTML output)
- ☐ Filename is `lab-XX-topic.qmd`, not `Untitled.qmd`